

TISP Nexus Submission Testing Report

Project: Implementing Security Measures within a Threat Intelligence Sharing Platform. Prepared for final academic submission evidence and demo readiness.

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DASHBOARD

Pass

Lint and production build completed.

CTI TOOLS

Pass

STIX export and threat report self-test completed.

INFRA CONFIG

Pass

Lab and production Compose configurations validated.

NAVIGATION

Pass

All dashboard modules opened from the sidebar.

Executive Summary

The codebase is ready for submission as a project package. The React dashboard has been redesigned into a cleaner SOC-style interface, the automated analysis tools generate normalized and anonymized intelligence outputs, and both the lab and production Docker Compose configurations validate successfully with required secrets externalized.

The project now includes a one-click local installer and an in-app Technology Blueprint module. This gives evaluators a direct path from download to working dashboard, plus a clear explanation of the technology stack, configuration levels, and demo story.

Live Docker deployment was not executed in this pass because it starts multiple local services and may pull large container images. The safe validation

completed here includes `docker compose --env-file .env.example config` and `docker compose -f docker-compose.yml -f docker-compose.prod.yml --env-file .env.prod.example config`. A live lab run can be performed after explicit approval.

One-Click Installation

Default Demo Path

Windows users can double-click

`Start-TISP-Nexus.bat` or run

`.\install.ps1`.

The installer checks Node.js/npm, installs dashboard packages, generates a local lab `.env`, runs CTI tool tests, builds the dashboard, starts the local command center, and opens

`http://127.0.0.1:5173`.

Advanced Modes

```
.\install.ps1 -Mode Validate
.\install.ps1 -Mode Lab
.\install.ps1 -Mode Lab -StartLabStack
.\install.ps1 -Mode Production
```

Technology Stack

LAYER	TECHNOLOGIES	PURPOSE
Command center	React 19, Vite, Tailwind CSS 4, Framer Motion, Lucide, Recharts	SOC-style dashboard for monitoring, reporting, and project demonstration.
CTI platform	MISP, OpenCTI, TheHive, STIX 2.1, TAXII-ready exports	Threat ingestion, relationship analysis, intelligence sharing, and response coordination.
Security controls	Keycloak, RBAC, MFA plan, TLS reverse proxy, AES-256 storage plan, anonymization	Identity, authorization, privacy, and secure sharing governance.
Infrastructure	Docker Compose, Nginx, PostgreSQL, Redis, Elasticsearch, RabbitMQ, MinIO	Reproducible lab stack and production-ready deployment overlay.
Evidence and automation	Node.js CTI tools, PowerShell validators, browser smoke tests, Edge screenshots/PDF	Repeatable proof that the project builds, validates, and covers the assignment requirements.

Validation Matrix

AREA	COMMAND / METHOD	EVIDENCE	STATUS
Dashboard lint	<code>npm run lint</code>	<code>submission-evidence/logs/01-dashboard-lint.log</code>	Passed
Dashboard production build	<code>npm run build</code>	<code>submission-evidence/logs/02-dashboard-build.log</code>	Passed
Automated analysis tools	<code>npm test</code>	<code>tisp-tools/out/stix-bundle.json</code> , <code>tisp-tools/out/threat-report.json</code>	Passed
Environment validation	<code>.\validate-env.ps1 .env.example -AllowPlaceholders</code>	<code>submission-evidence/logs/04-env-structure-validation.log</code>	Passed
Docker Compose config	<code>docker compose --env-file .env.example config</code>	<code>submission-evidence/logs/05-compose-config-validation.log</code>	Passed
Production environment validation	<code>.\validate-env.ps1 .env.prod.example -Profile prod -AllowPlaceholders</code>	<code>submission-evidence/logs/06-prod-env-structure-validation.log</code>	Passed
Production Compose overlay	<code>docker compose -f docker-compose.yml -f docker-compose.prod.yml --env-file .env.prod.example config</code>	<code>submission-evidence/logs/07-prod-compose-config-validation.log</code>	Passed
Production exposure guard	<code>.\validate-prod-compose.ps1 .env.prod.example</code>	<code>submission-evidence/logs/08-prod-compose-exposure-validation.log</code>	Passed
Container image manifests	<code>.\validate-images.ps1 .env.prod.example -Production</code>	<code>submission-evidence/logs/10-image-manifest-validation.log</code>	Passed
Sidebar navigation smoke test	<code>node scripts/browser-smoke.mjs</code> over local Vite app	<code>submission-evidence/logs/09-sidebar-navigation-smoke.json</code>	Passed

AREA	COMMAND / METHOD	EVIDENCE	STATUS
Full live Docker lab / production pilot	<code>docker compose up -d</code> or production overlay deploy	Requires real secrets, DNS, TLS certificates, and approval before starting services	Pending approval

Dashboard Evidence

TISP Nexus

Threat Intelligence Platform

OVERVIEW

Executive Dashboard

Technology Blueprint

Architecture Map

CTI OPERATIONS

MISP Ingestion

OpenCTI Graph

TheHive Cases

CONTROLS

Identity & RBAC

Automation

Lab Ready

Build, tools, config pass

Project settings

Submission build v1.0

TISP Command Center

Multi-organization CTI sharing platform

Submission build

Systems nominal Evidence validated

Covered

Platform Modules

8/8

+12.4%

Shared Indicators

42.8K

12 resolved

Open Cases

24

ISO/NIST mapped

Compliance Score

91%

Threat Intelligence Exchange Flow

MISP ingestion

OpenCTI correlation

TheHive response

n8n automation

Multi-organization sharing simulation

ORG-A SOC

ORG-B CERT

ORG-C DFIR

Priority Threat Stream

C2 domain cluster linked to ransomware activity

SSH brute-force IP range observed by two partners

Suspicious PowerShell YARA match submitted

Lab configuration validated 14 subsystems modeled TISP Nexus submission build

Desktop viewport: redesigned TISP Command Center with CTI exchange flow, threat stream, and submission status.



TISP Command Center

Multi-organization CTI sharing platform

● **Systems nominal** Evidence validated



Platform Modules

8/8



Shared Indicators

42.8K



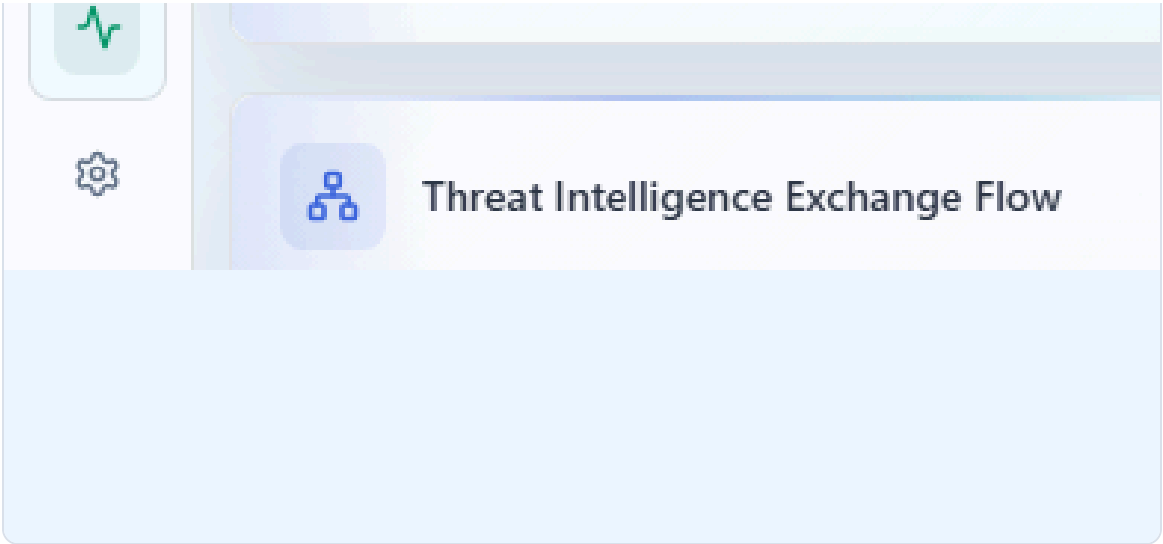
Open Cases

24

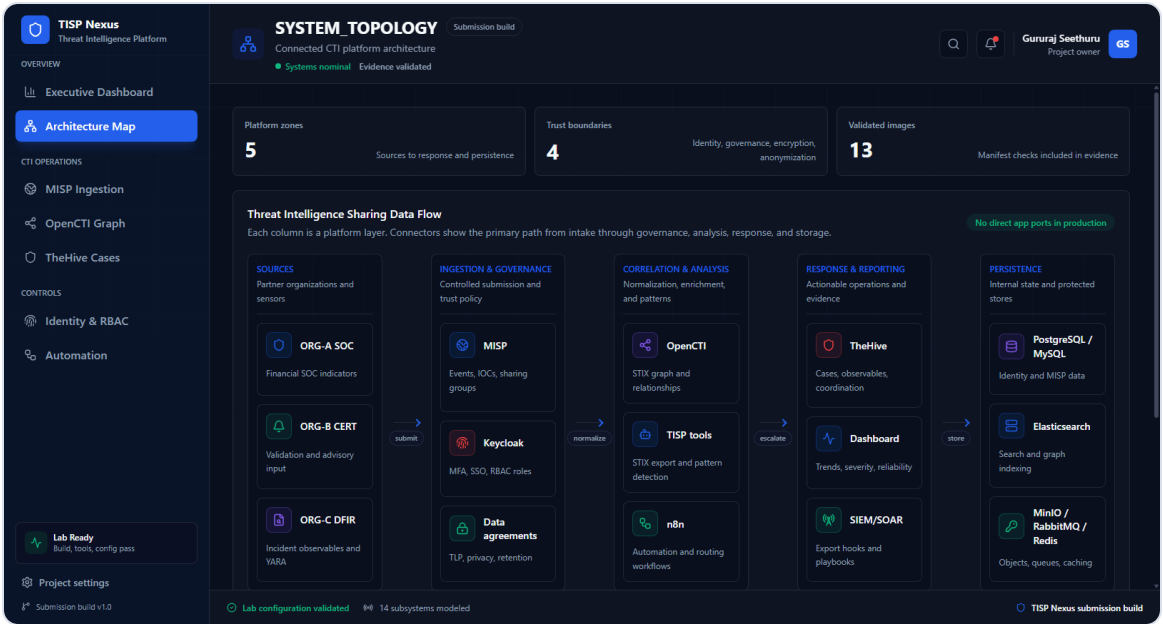


Compliance Score

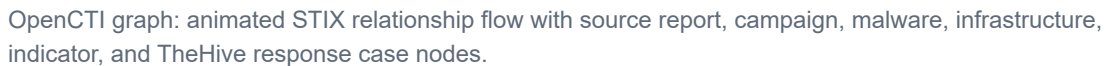
91%



Mobile viewport: responsive sidebar rail and single-column KPI layout without horizontal clipping.



Architecture map: connected CTI topology from sources through governance, analysis, response, and persistence layers.



Functional Navigation Evidence

SideBar Item	Expected View	Status
Architecture Map	SYSTEM_TOPOLOGY	Passed
MISP Ingestion	MISP_INTEL_MATRIX	Passed
OpenCTI Graph	OPEN_CTI_NEXUS	Passed
TheHive Cases	THE_HIVE_OPERATIONS	Passed
Identity & RBAC	IDENTITY_MESH	Passed
Automation	AUTOMATION_CONTROL	Passed
Executive Dashboard	TISP Command Center	Passed

Project Deliverables

DELIVERABLE	REPOSITORY EVIDENCE
Architecture and deployment documentation	Project 1/D1_Architecture_Deployment.docx , tisp-infra/docker-compose.yml
Data sharing agreement templates	Project 1/D2_Data_Sharing_Agreements.docx
Configuration guides	Project 1/D3_Configuration_Guides.docx , tisp-infra/.env.example
Automated analysis tools	tisp-tools/src/normalize-stix.mjs , tisp-tools/src/pattern-detection.mjs
Integration manuals	Project 1/D5_Integration_Manuals.docx
Reporting dashboards	tisp-command-center , dashboard screenshots in this report
Compliance documentation	Project 1/D7_Compliance_Documentation.docx , docs/PROJECT_COMPLETION_MATRIX.md
User training guides	Project 1/D8_User_Training_Guide.docx
Production operations	docs/PRODUCTION_READINESS_RUNBOOK.md , docs/PRODUCTION_ACCEPTANCE_CHECKLIST.md , docs/BACKUP_RESTORE_GUIDE.md

Conclusion

The project is suitable for academic submission after one final optional live-lab demonstration. The required written deliverables are present, the dashboard is buildable and visually polished, the analysis scripts produce output evidence, and the lab plus production infrastructure configurations are validated. Real production exposure should wait until the runbook's DNS, TLS, MFA, RBAC, backup, restore, and smoke-test gates are completed with live evidence.